time taken for B to travel 5 m = 5 / 4 = 1.25 m
(or speed of A relative to B = 4 – 0 = 4 m s^{-1}, time taken for A to travel 5 m = 5 / 4 = 1.25 s)

6 **D**

If the constant time interval is *T*, distance travelled by each successive bead is $s = \frac{1}{2}g(nT)^2$ where *n* is an integer.

time	*s*
0	0
T	$\frac{1}{2}gT^2$
2*T*	$\frac{1}{2}g(2T)^2 = 4\left(\frac{1}{2}gT^2\right)$
3*T*	$\frac{1}{2}g(3T)^2 = 9\left(\frac{1}{2}gT^2\right)$
nT	$\frac{1}{2}g(nT)^2 = n^2\left(\frac{1}{2}gT^2\right)$

7 **A**

$P = Fv$

$0.4 \times 800 = F \times 13$